

2020 级秋线性代数 1-A 参考答案

一、 计算题 (1-8 小题, 每小题 6 分, 共 48 分)

1. 设 $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} X + \begin{pmatrix} 2 & 5 \\ 0 & 3 \end{pmatrix} = 12 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, 求 X .

解: $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} X + \begin{pmatrix} 2 & 5 \\ 0 & 3 \end{pmatrix} = 12 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} X = 12 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} - \begin{pmatrix} 2 & 5 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 10 & 7 \\ 12 & 9 \end{pmatrix}$ 3'
 $\Rightarrow X = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}^{-1} \begin{pmatrix} 10 & 7 \\ 12 & 9 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}$ 3'

2. 计算四阶行列式 $\begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix}$

解: $\begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix} = \begin{vmatrix} x & x & 0 & 0 \\ 1 & 1-x & 1 & 1 \\ 0 & 0 & y & y \\ 1 & 1 & 1 & 1-y \end{vmatrix} = xy \begin{vmatrix} 1 & 1 & 0 & 0 \\ 1 & 1-x & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix}$ 3'
 $= xy \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix} = xy \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & -x & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -y \end{vmatrix} = xy \begin{vmatrix} -x & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & -y \end{vmatrix} = x^2 y^2$ 3'

3. 求 $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^{2020} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{2021}$.

解: $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^{2020} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{2021} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{2021} = \begin{pmatrix} 1 & 2+2021 \\ 3 & 4+6063 \end{pmatrix} = \begin{pmatrix} 1 & 2023 \\ 3 & 6067 \end{pmatrix}$ 3'

4. 设 A 是一个 3×4 矩阵, 其秩 $\text{rank}(A) = 2$, 方程组 $Ax = b$ 有三个特解

$\alpha = (1, 2, -1, 2)^T$, $\beta = (2, -1, 1, 3)^T$, $\gamma = (3, 2, -2, 1)^T$, 求其通解.

解: 由题意可知, 方程组 $Ax = 0$ 的基础解系含有两个线性无关的解向量.

$\beta - \alpha = (1, -3, 2, 1)^T$, $\gamma - \alpha = (2, 0, -1, -1)^T$ 为 $Ax = 0$ 的解, 且线性无关, 故可作为基础解系, 于是通解为

$$x = k_1 \begin{pmatrix} 1 \\ -3 \\ 2 \\ 1 \end{pmatrix} + k_2 \begin{pmatrix} 2 \\ 0 \\ -1 \\ -1 \end{pmatrix} + \begin{pmatrix} 3 \\ 2 \\ -2 \\ 1 \end{pmatrix}, k_1, k_2 \text{ 为任意实数。} \quad 2'$$

5. 求 $\mathbb{R}^3$ 中向量 $\alpha = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$ 在基 $\alpha_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, $\alpha_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, $\alpha_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ 下的坐标。

解: 设 $\alpha = x_1 \alpha_1 + x_2 \alpha_2 + x_3 \alpha_3$, 则 $2'$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = (\alpha_1 \alpha_2 \alpha_3)^{-1} \alpha = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \quad 4'$$

6. 在 $\mathbb{R}^2$ 中, 求从基 $\alpha_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\alpha_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ 到另一组基 $\beta_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, $\beta_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$ 的过渡矩阵。

解: 设 $\beta_1 = a\alpha_1 + b\alpha_2 = (\alpha_1, \alpha_2) \begin{pmatrix} a \\ b \end{pmatrix}$, $\beta_2 = c\alpha_1 + d\alpha_2 = (\alpha_1, \alpha_2) \begin{pmatrix} c \\ d \end{pmatrix}$, 即

$$(\beta_1, \beta_2) = (\alpha_1, \alpha_2) \begin{pmatrix} a & c \\ b & d \end{pmatrix}. \quad 3'$$

$$\text{过渡矩阵 } \begin{pmatrix} a & c \\ b & d \end{pmatrix} = (\alpha_1, \alpha_2)^{-1} (\beta_1, \beta_2) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 3 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ -1 & -1 \end{pmatrix} \quad 3'$$

7. 设 $\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & a \\ d & e & f \\ b & \frac{1}{\sqrt{2}} & c \end{pmatrix}$ 为正交矩阵, $f > 0$, 求 a, b, c, d, e, f .

解: 由第一行的平方和为 1 可得 $a = 0$,

$$\text{第一行于第三行点乘为 } 0: b \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + 0 \cdot c = 0 \Rightarrow b = -\frac{1}{\sqrt{2}}$$

第三行的平方和为 1 可得 $c = 0$, $3'$

由最后一列的平方和是 1 及 $f > 0$ 得 $f = 1$.

由第二行的平方和为 1 可得 $d = e = 0$ $3'$

即矩阵为
$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix}$$

8. 若二次型 $f(x_1, x_2, x_3) = 2x_1^2 + 3x_2^2 + tx_3^2 + 2x_1x_2 + 2x_1x_3$ 正定, 求 t 的取值范围。

解: 二次型的矩阵为
$$\begin{pmatrix} 2 & 1 & 1 \\ 1 & 3 & 0 \\ 1 & 0 & t \end{pmatrix}$$

它的三个顺序主子式为: $2, \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} = 5, \begin{vmatrix} 2 & 1 & 1 \\ 1 & 3 & 0 \\ 1 & 0 & t \end{vmatrix} = 5t - 3$

由二次型正定可得 $t > \frac{3}{5}$.

二、解答题 (9-11 小题, 每小题 12 分, 共 36 分)

9. 当 a, b 为何值时, 线性方程组
$$\begin{cases} x_1 + x_2 + x_3 + x_4 = 0 \\ x_1 + 2x_3 + 2x_4 = 1 \\ -x_1 - 2x_3 + (a-3)x_4 = b \\ 2x_1 + 3x_2 + ax_3 + x_4 = -1 \end{cases}$$
 无解, 有唯一解, 有无穷多解, 并在有无穷多解时求出通解。

解: 对方程组的增广矩阵进行初等行变换:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 2 & 2 & 1 \\ -1 & 0 & -2 & a-3 & b \\ 2 & 3 & a & 1 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & 1 & 1 & 1 \\ 0 & 1 & -1 & a-2 & b \\ 0 & 1 & a-2 & -1 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & a-2 & b \\ 0 & 1 & a-2 & -1 & -1 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & -1 & -1 \\ 0 & 0 & 0 & a-1 & b+1 \\ 0 & 0 & a-1 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & -1 & -1 \\ 0 & 0 & a-1 & 0 & 0 \\ 0 & 0 & 0 & a-1 & b+1 \end{pmatrix}$$

当 $a \neq 1$ 时, $\text{rank}(A|b) = \text{rank}(A) = 4$, 方程组有唯一解;

当 $a = 1, b \neq -1$ 时, $\text{rank}(A) = 2 < \text{rank}(A|b) = 3$, 方程组无解;

当 $a = 1, b = -1$ 时, $\text{rank}(A) = 2 = \text{rank}(A|b) = 2$, 方程组有无穷多解; 继续初等行变换:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

此时方程组的特解和导出组的基础解系为

$$\xi^* = \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \xi_1 = \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \xi_2 = \begin{pmatrix} -2 \\ 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

通解: $x = \xi^* + k_1 \xi_1 + k_2 \xi_2$, k_1, k_2 为任意常数.

10. 求向量组 $\alpha_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\alpha_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$, $\alpha_3 = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}$, $\alpha_4 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, $\alpha_5 = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ 的秩和极大无关

组, 并将其余向量用极大无关组表示.

$$\text{解: } \begin{pmatrix} 1 & 0 & 3 & 1 & 2 \\ 2 & 1 & 1 & 0 & 1 \\ 3 & 1 & 4 & 1 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 3 & 1 & 2 \\ 0 & 1 & -5 & -2 & -3 \\ 0 & 1 & -5 & -2 & -3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 3 & 1 & 2 \\ 0 & 1 & -5 & -2 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

所以 $\text{rank}(\alpha_1, \dots, \alpha_5) = 2$,

极大无关组可取为 α_1, α_2 .

$$\alpha_3 = 3\alpha_1 - 5\alpha_2,$$

$$\alpha_4 = \alpha_1 - 2\alpha_2,$$

$$\alpha_5 = 2\alpha_1 - 3\alpha_2,$$

11. 设矩阵 $A = \begin{pmatrix} 0 & 2 & -3 \\ -1 & 3 & -3 \\ 1 & -2 & 4 \end{pmatrix}$, 求可逆矩阵 P , 使 $P^{-1}AP$ 为对角矩阵.

解: 解特征方程 $|A - \lambda E| = 0$ 得特征值 $\lambda_1 = \lambda_2 = 1, \lambda_3 = 5$. 3'

解线性方程组 $(A - E)x = 0$ 得特征向量

$$\xi_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \xi_2 = \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix}. \quad \text{4'}$$

解线性方程组 $(A - 5E)x = 0$ 得特征向量

$$\xi_3 = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}. \quad \text{3'}$$

$$\text{令 } P = (\xi_1, \xi_2, \xi_3) = \begin{pmatrix} 2 & -3 & -1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{pmatrix}, \text{ 则 } P^{-1}AP = \begin{pmatrix} 1 & & \\ & 1 & \\ & & 5 \end{pmatrix}. \quad \text{2'}$$

三、证明题 (12-13 小题, 每小题 8 分, 共 16 分)

12. 已知向量组 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, 证明: 向量组 $\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1$ 线性无关,

而向量组 $\alpha_1 - \alpha_2, \alpha_2 - \alpha_3, \alpha_3 - \alpha_1$ 线性相关。

证明: 设 $k_1(\alpha_1 + \alpha_2) + k_2(\alpha_2 + \alpha_3) + k_3(\alpha_3 + \alpha_1) = 0$,

于是 $(k_1 + k_3)\alpha_1 + (k_1 + k_2)\alpha_2 + (k_2 + k_3)\alpha_3 = 0$ 。

由 $\alpha_1, \alpha_2, \alpha_3$ 线性无关可得 $k_1 + k_3 = 0, k_1 + k_2 = 0, k_2 + k_3 = 0$ 3'

而此方程组有唯一解 $k_1 = k_2 = k_3 = 0$, 故向量组 $\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1$ 线性无关。 2'

由 $(\alpha_1 - \alpha_2) + (\alpha_2 - \alpha_3) + (\alpha_3 - \alpha_1) = 0$, 可知 $\alpha_1 - \alpha_2, \alpha_2 - \alpha_3, \alpha_3 - \alpha_1$ 有一个非零的线

性组合为零, 组合系数为 1, 1, 1, 故向量组 $\alpha_1 - \alpha_2, \alpha_2 - \alpha_3, \alpha_3 - \alpha_1$ 线性相关。 3'

13. 设三阶矩阵 A 有三个不同的特征值 $\lambda_1, \lambda_2, \lambda_3$, 对应的特征向量为 $\alpha_1, \alpha_2, \alpha_3$, 令

$\beta = \alpha_1 + \alpha_2 + \alpha_3$, 证明: $\beta, A\beta, A^2\beta$ 线性无关.

证明: 依题意 $A\alpha_i = \lambda_i\alpha_i, i = 1, 2, 3$. 因此

$$A\beta = A(\alpha_1 + \alpha_2 + \alpha_3) = \lambda_1\alpha_1 + \lambda_2\alpha_2 + \lambda_3\alpha_3,$$

$$A^2\beta = \lambda_1^2\alpha_1 + \lambda_2^2\alpha_2 + \lambda_3^2\alpha_3,$$

设 $k_1\beta + k_2A\beta + k_3A^2\beta = 0$.

于是

$$(k_1 + k_2\lambda_1 + k_3\lambda_1^2)\alpha_1 + (k_1 + k_2\lambda_2 + k_3\lambda_2^2)\alpha_2 + (k_1 + k_2\lambda_3 + k_3\lambda_3^2)\alpha_3 = 0$$

由于不同特征值对应的特征向量是线性无关的, 从而

$$\begin{cases} k_1 + k_2\lambda_1 + k_3\lambda_1^2 = 0, \\ k_1 + k_2\lambda_2 + k_3\lambda_2^2 = 0, \\ k_1 + k_2\lambda_3 + k_3\lambda_3^2 = 0. \end{cases}$$

此方程组的系数行列式

$$\begin{vmatrix} 1 & \lambda_1 & \lambda_1^2 \\ 1 & \lambda_2 & \lambda_2^2 \\ 1 & \lambda_3 & \lambda_3^2 \end{vmatrix} = (\lambda_3 - \lambda_1)(\lambda_3 - \lambda_2)(\lambda_2 - \lambda_1) \neq 0,$$

从而方程组只有零解, 即 $k_1 = k_2 = k_3 = 0$, 从而 $\beta, A\beta, A^2\beta$ 线性无关.